

Department Of Computer Science
Gujarat University



Certificate

Roll no : 02(DS)

Seat no: 10002

This is to certify that Mr.siddharth Lakkad student of first of semester M.Sc. (Artificial Intelligence & Machine Learning) Defence specific s has duly completed his project titled health spy with blood pressure prediction for the semester ending in December 2022, towards partial fulfilment of his Degree of Master in Science (Artificial Intelligence & Machine Learning) Defence specific.

Date of Submission: 07/01/2023

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*This is to certify that **Mr. Raj Vasoya** student of first semester of M.Sc. (Artificial Intelligence & Machine Learning) Defence specific s has duly completed his project titled **Health Spy with blood pressure prediction** for the semester ending in December 2022, towards partial fulfilment of his Degree of Master in Science (Artificial Intelligence & Machine Learning) Defence specific.*

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Abstract

We are working on health spy. In this project we calculate a BMI & Water intake and also predicting a blood pressure prediction of the user can get the result of the blood pressure, water intake and also BMI (Body mass index) **We use 13 parameter of find a blood pressure like blood pressure abnormality, Level of hemoglobin, genetic pedigree coefficient, age, BMI, Sex, Pregnancy, Smoking, Physical Activity, Salt content In the diet, Alcohol consumption per day, Level of Stress, chronicle kidney disease Etc.**

INTRODUCTION

The high risk attributed to obesity is related particularly to its association with increased risk factors for cardiovascular disease, notably hypertension. In order to adopt early preventive/therapeutic actions to minimize the risk of cardiovascular events in obese individuals, methods that can predict hypertension using low cost procedures are necessary, especially in underdeveloped and in developing countries.

Previous studies have proven that incorporating physics-based knowledge into a machine learning model can encourage the model to make physically consistent and more accurate predictions. Additionally, literature shows that relevant knowledge may be valuable when dealing with insufficient amounts of data. Based on previous studies of IML, This project implements two approaches for IML prediction of blood pressure curves. The first method incorporates a physics-based loss function, and the second method predicts the residual of a physics model's estimate.

The medical condition of hypertension is defined as having a persistently blood pressure above 140/90mmHg. The high pressures put an extra load on the vessels, heart, and organs such as the brain and kidneys, and cause a higher risk for lifestyle diseases like heart failure, stroke, and kidney disease (Chobanian et al., 2003a). The World Health Organization (WHO) estimates that more than one billion people have hypertension, and the majority are unaware of the condition as the symptoms may not be present (WHO). Hypertension is the cause of around 9 million deaths every year (Halдар, 2013), and WHO considers it a major cause for premature deaths worldwide.

National Center for Health Statistics in the United States documented that more than half of the individuals treated with hypertension medication between 2013 and 2016 were still hypertensive in 2017 (CDC). A possible reason for this is incorrect medication as common guidelines often are used in treatment. Objective guidelines tend to draw sharp thresholds for treatment, ignoring the multiple interconnected risk factors which provide the need for an individualized treatment (Eddy et al., 2011).

Blood pressure and hypertension:

Blood pressure is a result of the blood being pumped from the heart and pushed against the walls of the blood vessels. The highest pressure is achieved during.

Body Mass Index (BMI):

BMI is a person's weight in kilograms (or pounds) divided by the square of height in meters (or feet). A high BMI can indicate high body fatness. BMI screens for weight categories that may lead to health problems, but it does not diagnose the body fatness or health of an individual.

Water intake:

Getting enough water every day is important for your health. Drinking water can prevent dehydration, a condition that can cause unclear thinking, result in mood change, cause your body to overheat, and lead to constipation and kidney stones. The U.S. National Academies of Sciences, Engineering, and Medicine determined that an adequate daily fluid intake is: About 15.5 cups (3.7 liters) of fluids a day for men. About 11.5 cups (2.7 liters) of fluids a day for women.

Short information about method

Method for predicting blood pressure

Using Machine learning we predict blood pressure in machine learning we used classification method. In a classification using logistic regression .And use the decision tree classifier algorithm to checking blood pressure from of array.

Calculate BMI :

With the metric system, the formula for BMI is weight in kilograms divided by height in meters squared. Since height is commonly measured in centimetres, an alternate calculation formula, dividing the weight in kilograms by the height in centimetres squared, and then multiplying the result by 10,000, can be used.

Formula:

$$\begin{aligned} & \text{[Weight (kg) / [height (m)]}^2 \\ & \text{Or} \\ & \text{[Weight (kg) / height (cm) / height (cm)] x 10,000} \end{aligned}$$

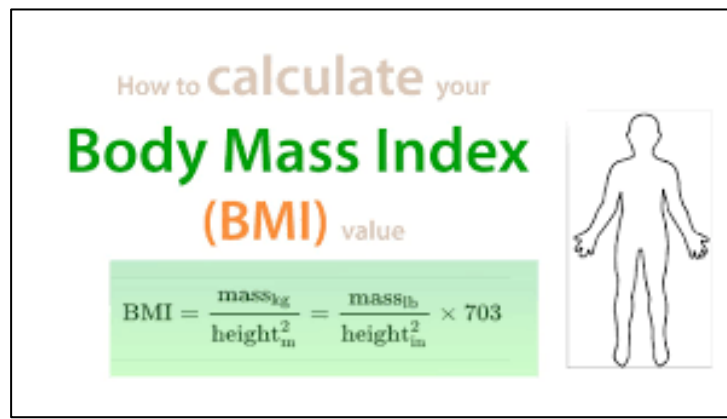


Figure 1

Water Intake :

This calculator helps you determine approximately the amount of water you must drink each day to avoid dehydration. Remember, this is only an estimate. The actual amount one needs depends on many factors like exercise, illness and fluid content in dietary intake. Pregnant and breast feeding women need to drink additional fluids to stay hydrated

Formula uses:

- For 0 - 10 kg = weight (kg) x 100 mL/kg/day
- For 10-20 kg = 1000 mL + [weight (kg) x 50 ml/kg/day]
- For > 20 kg = 1500 mL + [weight (kg) x 20 ml/kg/day]

Note: Infusion rate = total fluid volume per day ÷ 24 hour

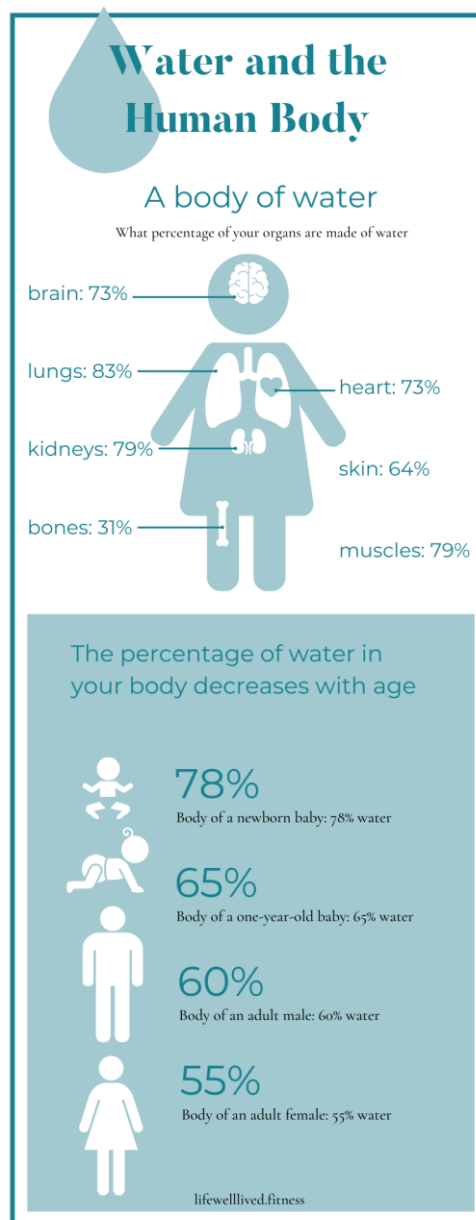


Figure 2

About Machine Learning Concepts & Algorithm to Train Modal

Classification

Classification is a technique to categorize data into desired and distinct number of classes where we can assign label to each class Example: Email- spam or not, Animal- cat or dog It solves binary problems whether the value is yes or no, that is whether email is spam or not

Logistic Regression

- Logistic regression is one of the most popular Machine Learning algorithms, which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables.
- Logistic regression predicts the output of a categorical dependent variable. Therefore the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, **it gives the probabilistic values which lie between 0 and 1.**
- Logistic Regression is much similar to the Linear Regression except that how they are used. Linear Regression is used for solving Regression problems, whereas **Logistic regression is used for solving the classification problems.**
- In Logistic regression, instead of fitting a regression line, we fit an "S" shaped logistic function, which predicts two maximum values (0 or 1).
- The curve from the logistic function indicates the likelihood of something such as whether the cells are cancerous or not, a mouse is obese or not based on its weight, etc.
- Logistic Regression is a significant machine learning algorithm because it has the ability to provide probabilities and classify new data using continuous and discrete datasets.
- Logistic Regression can be used to classify the observations using different types of data and can easily determine the most effective variables used for the classification. The below image is showing the logistic function:

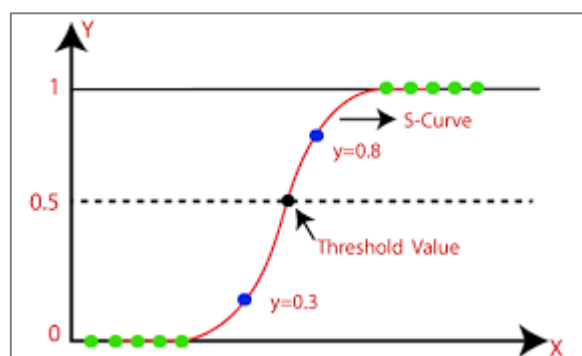


Figure 3

Logistic Function

- The sigmoid function is a mathematical function used to map the predicted values to probabilities.
- It maps any real value into another value within a range of 0 and 1.
- The value of the logistic regression must be between 0 and 1, which cannot go beyond this limit, so it forms a curve like the "S" form. The S-form curve is called the Sigmoid function or the logistic function.
- In logistic regression, we use the concept of the threshold value, which defines the probability of either 0 or 1. Such as values above the threshold value tends to 1, and a value below the threshold values tends to 0.

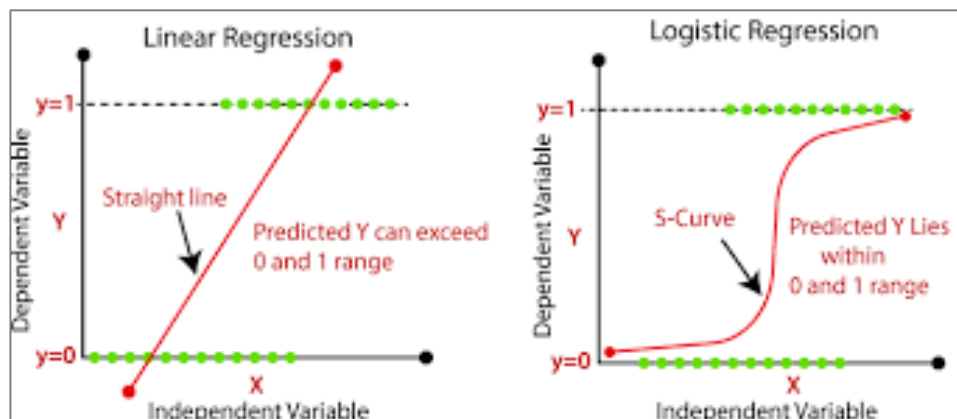


Figure 4

Assumptions for Logistic Regression:

- The dependent variable must be categorical in nature.
- The independent variable should not have multi-collinearity.

Logistic Regression Equation:

The Logistic regression equation can be obtained from the Linear Regression equation. The mathematical steps to get Logistic Regression equations are given below: We know the equation of the straight line can be written as:

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \dots + b_nx_n$$

In Logistic Regression y can be between 0 and 1 only, so for this let's divide the above equation by $(1-y)$:

$$\frac{y}{1-y}; 0 \text{ for } y=0, \text{ and infinity for } y=1$$

But we need range between $-\text{[infinity]}$ to $+\text{[infinity]}$, then take logarithm of the equation it will become:

$$\log \left[\frac{y}{1-y} \right] = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \dots + b_nx_n$$

The above equation is the final equation for Logistic Regression.

Decision Tree Classification Algorithm

- Decision Tree is a **supervised learning technique** that can be used for both classification and Regression problems, but mostly it is preferred for solving Classification problems. It is a tree-structured classifier, where **internal nodes represent the features of a dataset, branches represent the decision rules** and **each leaf node represents the outcome**.
- In a Decision tree, there are two nodes, which are the **Decision Node** and **Leaf Node**. Decision nodes are used to make any decision and have multiple branches, whereas Leaf nodes are the output of those decisions and do not contain any further branches.
- The decisions or the test are performed on the basis of features of the given dataset.
- ***It is a graphical representation for getting all the possible solutions to a problem/decision based on given conditions.***
- It is called a decision tree because, similar to a tree, it starts with the root node, which expands on further branches and constructs a tree-like structure.
- In order to build a tree, we use the **CART algorithm**, which stands for **Classification and Regression Tree algorithm**.
- A decision tree simply asks a question, and based on the answer (Yes/No), it further split the tree into subtrees.
- Below diagram explains the general structure of a decision tree

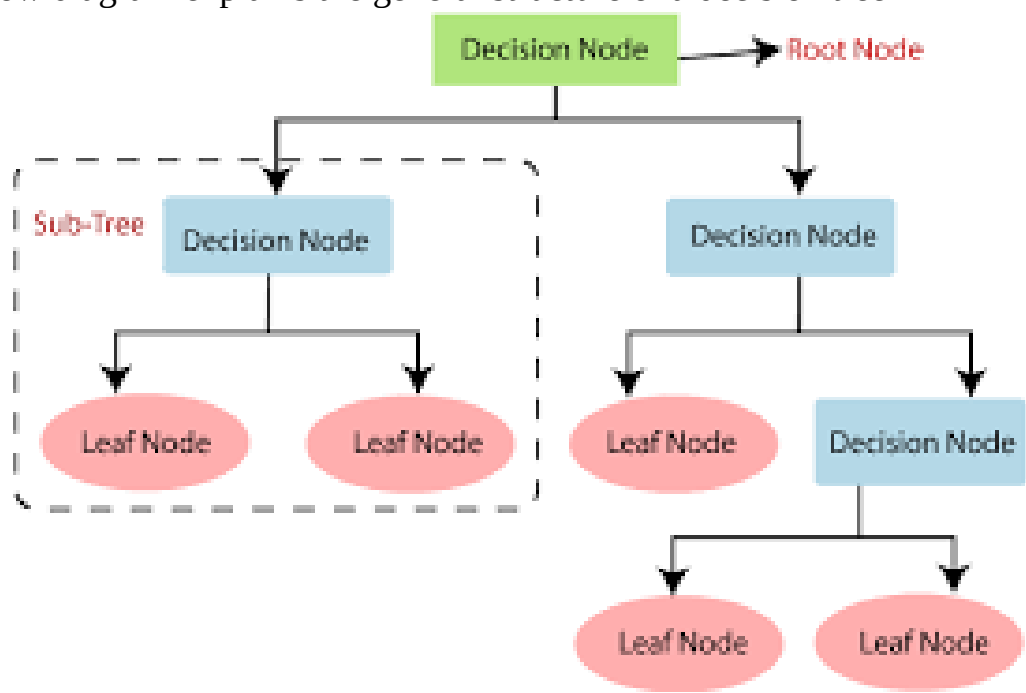


Figure 5

Why use Decision Trees?

There are various algorithms in Machine learning, so choosing the best algorithm for the given dataset and problem is the main point to remember while creating a machine learning model. Below are the two reasons for using the Decision tree

- Decision Trees usually mimic human thinking ability while making a decision, so it is easy to understand.

- The logic behind the decision tree can be easily understood because it shows a tree-like structure.

Decision Tree Terminologies:

- **Root Node:** Root node is from where the decision tree starts. It represents the entire dataset, which further gets divided into two or more homogeneous sets.
- **Leaf Node:** Leaf nodes are the final output node, and the tree cannot be segregated further after getting a leaf node.
- **Splitting:** Splitting is the process of dividing the decision node/root node into sub-nodes according to the given conditions.
- **Branch/Sub Tree:** A tree formed by splitting the tree.
- **Pruning:** Pruning is the process of removing the unwanted branches from the tree.
- **Parent/Child node:** The root node of the tree is called the parent node, and other nodes are called the child nodes.

How does the Decision Tree algorithm Work?

In a decision tree, for predicting the class of the given dataset, the algorithm starts from the root node of the tree. This algorithm compares the values of root attribute with the record (real dataset) attribute and, based on the comparison, follows the branch and jumps to the next node.

For the next node, the algorithm again compares the attribute value with the other sub-nodes and move further. It continues the process until it reaches the leaf node of the tree. The complete process can be better understood using the below algorithm:

Step-1: Begin the tree with the root node, says S, which contains the complete dataset.

Step-2: Find the best attribute in the dataset using Attribute Selection Measure (ASM).

Step-3: Divide the S into subsets that contains possible values for the best attributes.

Step-4: Generate the decision tree node, which contains the best attribute.

Step-5: Recursively make new decision trees using the subsets of the dataset created in step -3. Continue this process until a stage is reached where you cannot further classify the nodes and called the final node as a leaf node.

Advantages of the Decision Tree

- It is simple to understand as it follows the same process which a human follow while making any decision in real-life.
- It can be very useful for solving decision-related problems.
- It helps to think about all the possible outcomes for a problem.
- There is less requirement of data cleaning compared to other algorithms.

Brief Description about Relation between Blood Pressure and Using Parameters

Relation between Hemoglobin and blood pressure

It has been hypothesized that an increased hemoglobin level elevates blood pressure. The present study investigated the association between hemoglobin level and systolic blood pressure and diastolic blood pressure in healthy persons. The study population was composed of 101 377 whole blood and plasma donors, who made 691 107 visits to the blood bank. At each visit, hemoglobin level and blood pressure were measured as part of the standard procedure before a blood donation. We used repeated measurement analysis to analyze the data.

We used generalized estimating equation models to assess the between-person effect and linear mixed models to assess the within-person effect. All of the analyses were done separately for men and women. In the study population, 50% were men. The mean age in men was 49.3 years (± 12.5 years), and in women it was 42.4 years (± 13.7 years). Hemoglobin level was positively associated with both systolic and diastolic blood pressures. With respect to the between-person effect, regression coefficients for systolic blood pressure were 1.3 mm Hg per millimole per liter increase in hemoglobin level for men and 1.8 mm Hg per millimole per liter increase in hemoglobin level for women. With respect to the within-person effect, regression coefficients for systolic blood pressure were 0.7 mm Hg and 0.9 mm Hg per millimole per liter increase in hemoglobin level for men and women, respectively. For diastolic blood pressure, results were comparable. The results show that hemoglobin level is positively associated with both systolic and diastolic blood pressures in healthy individuals. We observed consistent effects between persons but also within persons.

Relation between Genetic Pedigree Coefficient and blood pressure

The role of heredity in influencing blood pressure and risk of hypertension is well recognized. However, progress in identifying specific genetic variation that contributes to heritability is very limited. This is in spite of completion of the human genome sequence, the development of extraordinary amounts of information about genome sequence variation and the investigation of blood pressure inheritance in linkage analysis, candidate gene studies and, most recently genome-wide association studies. This paper considers the progress of this research and the obstacles that have been encountered.

This work has made clear that the genetic architecture of blood pressure regulation in the population is not likely to be shaped by commonly occurring genetic variation in a discrete set of blood pressure-influencing genes. Rather heritability may be accounted for by rare variation that has its biggest impact within pedigrees rather than on the population at large. Rare variants in a wide range of genes are likely to be the focus of high blood pressure genetics for the next several years and the emerging strategies that can be applied to uncover this genetic variation and the problems that must confronted are considered.

Relation between Age and blood pressure

Blood pressure ranges for adults are:

- **High:** Systolic (top number) of 130 or above and/or diastolic (bottom number) of 80 or above
 - [High Blood Pressure](#) Stage 1: Systolic of 130-139 or diastolic of 80-89
 - [High Blood Pressure](#) Stage 2: Systolic of 140 or higher or diastolic 90 or higher
- **Elevated:** Systolic (top number) between 120 and 129 and diastolic (bottom number) of 79 or below
- **Normal:** Systolic (top number) of 119 or below and diastolic (bottom number) of 79 or below
- **Low:** Systolic (top number) of less than 100 and/or diastolic (bottom number) of less than 60

Some people may have normal or asymptomatic blood pressures lower than 100/60

In people who have chronically elevated blood pressure ([hypertension](#)), symptoms of [low blood pressure](#) may occur at readings above 100/60

Hypersensitive crisis: Systolic of 180 or higher and/or diastolic of 120 or higher. The normal blood pressure for adolescents 13 years or older is less than 120/80 mmHg. In younger children, the normal range for blood pressure is determined by the child's sex, age, and height. The normal range is expressed as a percentile, similar to charts used to track children's growth.

Blood pressure is separated into three categories based upon the child's blood pressure percentile:

- **High blood pressure (hypertension):** Either systolic and/or diastolic blood pressure 95th percentile or greater measured on three or more separate occasions, or if blood pressure exceeds 130/80 mmHg.
- **Elevated blood pressure:** Systolic and/or diastolic blood pressure is in the 90th percentile or greater, but less than the 95th percentile, or if blood pressure exceeds 120/80 mmHg (even if less than the 90th percentile for age, gender, and height).

BLOOD PRESSURE CATEGORY	SYSTOLIC mm Hg (upper number)		DIASTOLIC mm Hg (lower number)
NORMAL	LESS THAN 120	and	LESS THAN 80
ELEVATED	120 – 129	and	LESS THAN 80
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1	130 – 139	or	80 – 89
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2	140 OR HIGHER	or	90 OR HIGHER
HYPERTENSIVE CRISIS (consult your doctor immediately)	HIGHER THAN 180	and/or	HIGHER THAN 120

Figure 6

Relation between BMI and blood pressure

Change in body mass index and its impact on blood pressure: a prospective population study

Background: Overweight and obesity increase the risk of elevated blood pressure, but the knowledge of the effect of weight change on blood pressure is sparse.

Objective: To investigate the association between change in body mass index (BMI) and change in diastolic blood pressure (DBP), systolic blood pressure (SBP), and hypertension status.

Design: Two population-based cross-sectional studies, one in 1984-86 and the other in 1995-97.

Setting: The Nord-Trondelag Health Study (HUNT).

Participants: We included 15,971 women and 13,846 men who were 20 y or older at the first survey, without blood pressure medication at both surveys and without diabetes, cardiovascular disease or dysfunction in daily life at baseline.

Measurements: Weight, height and blood pressure were measured standardised. Change in BMI was categorised as stable (initial BMI $\pm$ 0.1 kg/m² each follow-up year), increased or decreased, and BMI was categorised by using World Health Organisation's categorisation (underweight BMI: <18.5 kg/m², normal weight BMI: 18.5-24.9 kg/m², overweight BMI: 25.0-29.9 kg/m², obesity BMI $\geq$ 30 kg/m²).

Results: An increase in BMI and a decrease in BMI were significantly associated with increased and decreased SBP and DBP, respectively, compared to a stable BMI in both genders and all age groups, although the strongest effect was found among those who were 50 y and older. The adjusted odds ratio for having hypertension at HUNT 2 was 1.8 (95% confidence interval (CI): 1.5, 2.2) among women and 1.6 (95% CI: 1.4, 1.8) among men aged 20-49 y who increased their BMI compared to those who had stable BMI. A similar, but weaker association was found among women and men aged 50 y or more. The mean change in both SBP and DBP was higher for those who changed BMI category from first to the second survey than for those who were in the same BMI class at both surveys.

Conclusions: Our result supports an independent effect of change in BMI on change in SBP and DBP in both women and men, and that people who increase their BMI are at increased risk for hypertension.

Gender differences in the regulation of blood pressure

Men are at greater risk for cardiovascular and renal disease than are age-matched, premenopausal women. Recent studies using the technique of 24-hour ambulatory blood pressure monitoring have shown that blood pressure is higher in men than in women at similar ages. After menopause, however, blood pressure

increases in women to levels even higher than in men. Hormone replacement therapy in most cases does not significantly reduce blood pressure in postmenopausal women, suggesting that the loss of estrogens may not be the only component involved in the higher blood pressure in women after menopause. In contrast, androgens may decrease only slightly, if at all, in postmenopausal women. In this review the possible mechanisms by which androgens may increase blood pressure are discussed.

Findings in animal studies show that there is a blunting of the pressure-natriuresis relationship in male spontaneously hypertensive rats and in ovariectomized female spontaneously hypertensive rats treated chronically with testosterone. The key factor in controlling the pressure-natriuresis relationship is the renin-angiotensin system (RAS). The possibility that androgens increase blood pressure via the RAS is explored, and the possibility that the RAS also promotes oxidative stress leading to production of vasoconstrictor substances and reduction in nitric oxide availability is proposed.

Effects of pregnancy on blood pressure

If you're pregnant, your doctor or nurse will likely check your blood pressure at every prenatal visit.

[Blood pressure](#) is the force of your blood as it pushes against artery walls while your heart pumps. It can go up or down at certain times of the day, and it may change if you're feeling excited or nervous.

Your blood pressure reading reveals important information about the health of you and your baby. It can also be a way for your doctor to determine if you have another condition that needs to be checked out, like [preeclampsia](#).

Changes that happen in your body during pregnancy can affect your blood pressure. When carrying a baby, your circulatory system expands quickly, which may cause a drop in blood pressure.

It's [common](#) [Trusted Source](#) for your blood pressure to lower in the first 24 weeks of pregnancy.

Other factors that can contribute to low blood pressure include:

- dehydration
- anaemia
- internal bleeding
- prolonged bed rest
- certain medications
- heart conditions
- endocrine disorders
- kidney disorders
- infections
- nutritional deficiencies

How smoking affects blood pressure

The addictive effects of smoking are only partly known, but it is likely that hemodynamic effects of tobacco smoking may contribute to the habituation. It has since long been known that blood pressure and heart rate increase during smoking. These effects are specifically associated with nicotine while the other components of which more than a thousand have been isolated seem to be of minor importance. The rise in blood pressure is due both to an increase in cardiac output and total peripheral vascular resistance. The blood pressure rise appears immediately and occurs before any increase in circulating catecholamine's. In hypertensive patients the blood pressure lowering effect of beta-blockers may be partly abolished by tobacco smoking whereas alpha-receptor blockers seem to maintain the antihypertensive efficacy in smokers. It is a paradox that while smoking acutely increases blood pressure, a slightly lower blood pressure level has been found among smokers than non-smokers in larger epidemiological studies. Because blood pressure may increase after cessation of smoking, a smoke quitting program should not postpone initiation of antihypertensive treatment in patients otherwise in need of such treatment.

Effects of exercise on blood pressure

Aerobic activities such as swimming, cycling, and running put additional demands on your cardiovascular system. Your muscles need more oxygen than they do when you're at rest, so you have to breathe more quickly.

Your heart starts to pump harder and faster to circulate blood to deliver oxygen to your muscles. As a result, systolic blood pressure rises.

It's normal for systolic blood pressure to rise to between 160 and 220 mm Hg during exercise. Unless you've cleared it with your doctor, stop exercising if your systolic blood pressure surpasses 200 mm Hg. Beyond 220 mm Hg, your risk of a heart problem increases.

Different factors can influence how your cardiovascular system responds to exercise. Some of these factors include diet, medical conditions, and medications.

For instance, exercise hypertension is a condition that causes an extreme spike in blood pressure during physical activity. People with exercise hypertension can experience spikes in systolic blood pressure up to 250 mm Hg during exercise.

In general, your blood pressure should return to normal within several hours of a workout. Even then, you might notice that your blood pressure doesn't return to exactly what it was before exercise. That's because it's normal for blood pressure to drop slightly within a few hours of exercise.

Regular physical activity **makes your heart stronger**. A stronger heart can pump more blood with less effort. If your heart can work less to pump, the force on your arteries decreases, **lowering your blood pressure**.

How does salt increase our blood pressure?

Normally our kidneys do a good job at regulating the sodium and water levels of our blood. However, for many of us, eating too much salt can disturb this balance, causing sodium levels in the blood to increase. This leads our body to hold onto more water and increases both the fluid surrounding our cells and the volume of blood in our bloodstream. As blood volume increases, the pressure on our blood vessels begins to increase and our heart needs to work harder to move blood around our body. Over time this extra strain can lead to stiffening of blood vessels and increases the risk of high blood pressure, heart disease and stroke

Alcohol consumption and blood pressure

The relationship between alcohol consumption and systolic and diastolic blood pressure was examined in a representative population sample of 1,429 men and women aged 35-64 years in Auckland, New Zealand during 1982. Univariate plots demonstrated a U-shaped relationship between alcohol consumption and systolic and diastolic blood pressures in men and in women aged 50 years and older, with light and moderate drinkers (less than 0-34 g alcohol/day) having lower blood pressure levels than either non-drinkers or heavy drinkers. No clear relationship was seen in younger women. The U-shaped relationship in men, particularly for systolic blood pressure, remained after controlling for potential confounders, whereas in women aged 50 years and older, there appeared to be a threshold level of approximately four drinks per day, below which drinkers had similar blood pressure levels to non-drinkers. In the multivariate analyses, it was calculated that among men, light and moderate drinkers had systolic levels 4.8 mmHg lower and diastolic levels 1.7 mmHg lower than non-drinkers and heavy drinkers, while among women aged 50-64 years, non-drinkers, light drinkers, or moderate drinkers had systolic levels 10.2 mmHg lower and diastolic levels 4.5 mmHg lower than heavy drinkers. These findings suggest that in men and in women aged 50 years and older, there is a nonlinear relationship between blood pressure and alcohol consumption, and that there is a level of alcohol consumption, of approximately four drinks per day, below which drinkers have either similar or lower blood pressure levels compared to non-drinkers.

Reactions to stress can affect blood pressure

The body releases a surge of hormones when under stress. These hormones cause the heart to beat faster and the blood vessels to narrow. These actions increase blood pressure for a time.

There's no proof that stress by itself causes long-term high blood pressure. But reacting to stress in unhealthy ways can raise blood pressure and increase the risk of heart attack and stroke. Behaviours linked to higher blood pressure include:

- Drinking too much alcohol or caffeine.

- Eating unhealthy foods.
- Eating too much.
- Not moving enough.

Heart disease also might be linked to certain health conditions related to stress, such as:

- Anxiety.
- Depression.
- Being cut off from friends and family.

There's no proof that these conditions are directly linked to high blood pressure. But the hormones the body makes when under emotional stress might damage arteries. The artery damage might lead to heart disease. And symptoms of depression and anxiety might cause some people to forget to take medicines to control high blood pressure or other heart conditions.

Stress can cause a steep rise in blood pressure. But when stress goes away, blood pressure returns to what it was before the stress. However, short spikes in blood pressure can cause heart attacks or strokes and may also damage blood vessels, the heart and the kidneys over time. The damage is like the damage from long-term high blood pressure.

Stress-reducing activities can help lower blood pressure

Although people with high stress and high blood pressure would generally see blood pressure go down after controlling stress, reducing stress might not lower blood pressure in everyone. But managing stress can help improve health in other ways. Learning how to manage stress can lead to healthy behaviour changes — including those that lower blood pressure.

Thyroid hormone and blood pressure regulation

Thyroid hormone has well-recognized effects on the cardiovascular system and blood pressure regulation. Blood pressure is altered across the entire spectrum of thyroid disease. The effects of hyperthyroidism include increased cardiac output, contractility, tachycardia, widened pulse pressure, decreased systemic vascular resistance, and increased basal metabolic rate. The manifestations of hypothyroidism are in marked contrast to those of hyperthyroidism and include decreased cardiac output, narrow pulse pressure, increased systemic vascular resistance, and decreased metabolic rate. Although thyroid hormone affects almost all tissues of the body and mediates changes in homeostasis, adaptations of the cardiovascular system can result in changes in blood pressure to accommodate the new demands on the system. In this paper, we review the direct and indirect thyroid hormone-mediated effects on blood pressure.

What is adrenal fatigue and blood pressure?

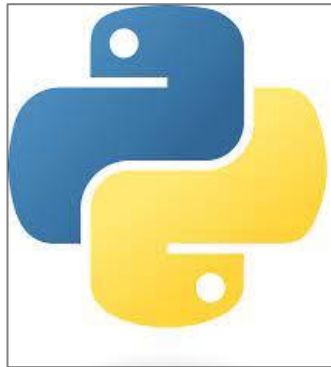
Blood pressure is an important indicator of adrenal health and function. Mild adrenal weakness is usually accompanied by normal to high blood pressure symptoms. As Adrenal Fatigue advances, low blood pressure, at rest or related to posture becomes more prevalent.

What is the relationship between high blood pressure and kidney disease?

High Blood Pressure and Chronic Kidney Disease. HBP and chronic kidney disease (CKD) are closely related. In 2013, high blood pressure led to more than 33,000 new cases of kidney failure, making it the second leading cause of kidney failure behind diabetes. More than 20% of people aged 20 years or older with hypertension have CKD.

About Used Technology

PYTHON



Python is a high-level, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation.

Python is dynamically-typed and garbage-collected. It supports multiple programming paradigms, including structured (particularly procedural), object-oriented and functional programming. It is often described as a "batteries included" language due to its comprehensive standard library

My SQL



MySQL is the world's most popular open source database. According to DB-Engines, MySQL ranks as the second-most-popular database, behind Oracle Database. MySQL powers many of the most accessed applications, including Facebook, Twitter, Netflix, Uber, Airbnb, Shopify, and Booking.com.

Since MySQL is open source, it includes numerous features developed in close cooperation with users over more than 25 years. So it's very likely that your favorite application or programming language is supported by MySQL Database.

About used library

Pandas:

Pandas library for Data analysis of data.

- Pandas is a python-based package that includes fast, flexible and expressive data structures, which are designed to work with both relational or labeled data. It is a fundamental high-level building block for doing practical, real-world data analysis in Python.

- Pandas is a library for Python, which has taken the data analysis world by storm.

- If anyone wants to get started with Machine Learning, Data analysis, or any data-intensive work, then Pandas should be the first thing to learn.

- It works well with the other libraries in Python like Numpy and Matplotlib, which are also essential in the field of data science.

- Reading Excel sheets full of data or any kind of data becomes super easy with pandas.



```
C: > Users > LENOVO > Downloads > project old.ipynb > import pandas as pd
+ Code + Markdown | Run All | Clear Outputs of All Cells | Outline ...

import pandas as pd
[2]

df = pd.read_csv("/content/Blood Pressure.csv")
[5]

df
[ ]
```

Figure 7

Sklearn

What is Scikit-Learn?

Scikit-learn (Sklearn) is the most useful and robust library for machine learning in Python. It provides a selection of efficient tools for machine learning and statistical modeling including classification, regression, clustering and dimensionality reduction via a consistent interface in Python. This library, which is largely written in Python, is built upon NumPy, SciPy and Matplotlib.


```

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split

le=LabelEncoder()

le
LabelEncoder()

bp_train, bp_test=train_test_split(df, test_size=0.2)

bp_train_x=bp_train.iloc[:, 1:-1]
bp_test_x=bp_test.iloc[:, 1:-1]

```

Figure 8

Pickle

Pickle in Python is primarily used in serializing and deserializing a Python object structure. In other words, it's the process of converting a Python object into a byte stream to store it in a file/database, maintain program state across sessions, or transport data over the network.

```

import pickle

pickle.dump(dtc, open("health.pickle", 'wb'))

loaded_model = pickle.load(open("/content/health.pickle", 'rb'))
result = loaded_model.score(bp_test_x, bp_test_y)
print(result)

0.835

```

Figure 9

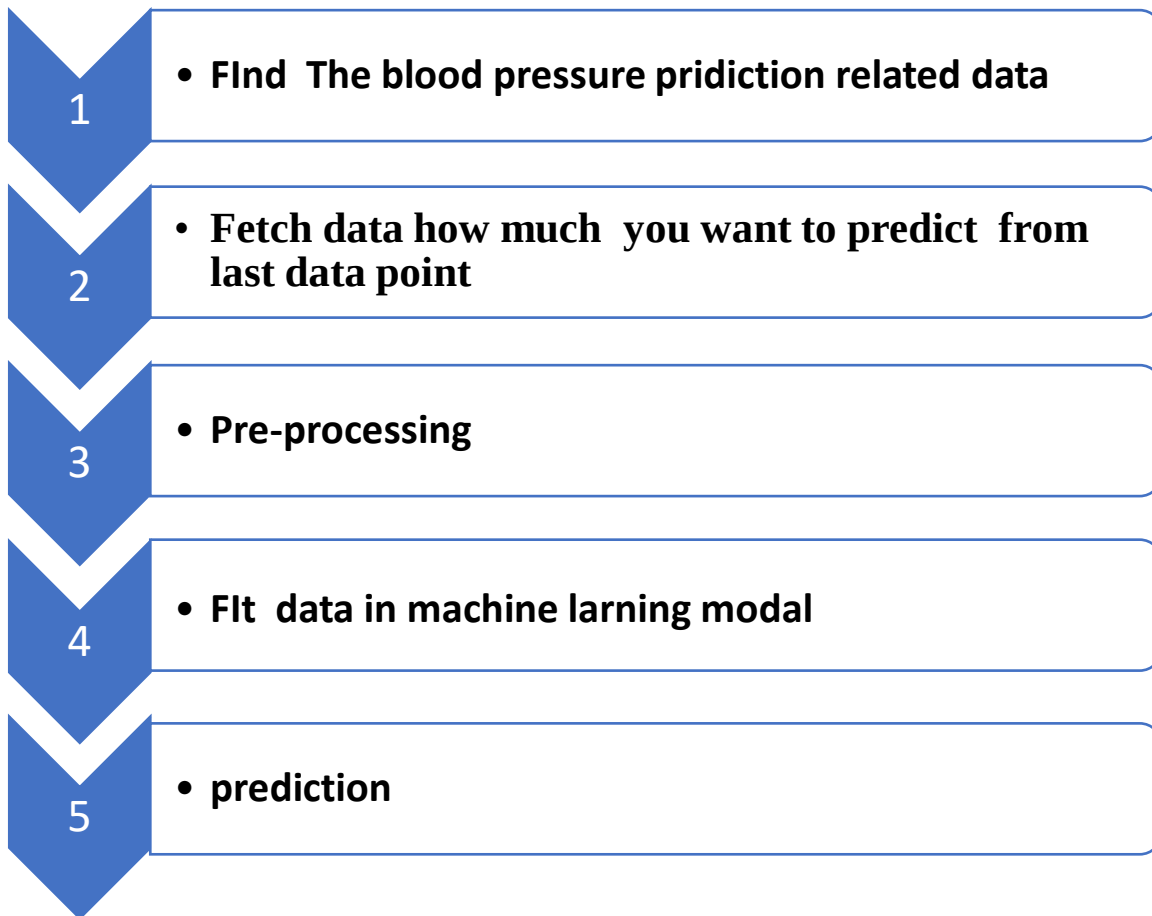
Tkinter

Tkinter is the standard GUI library for Python. Python when combined with Tkinter provides a fast and easy way to create GUI applications. Tkinter provides a powerful object-oriented interface to the Tk GUI toolkit. Import the Tkinter module.

PIL

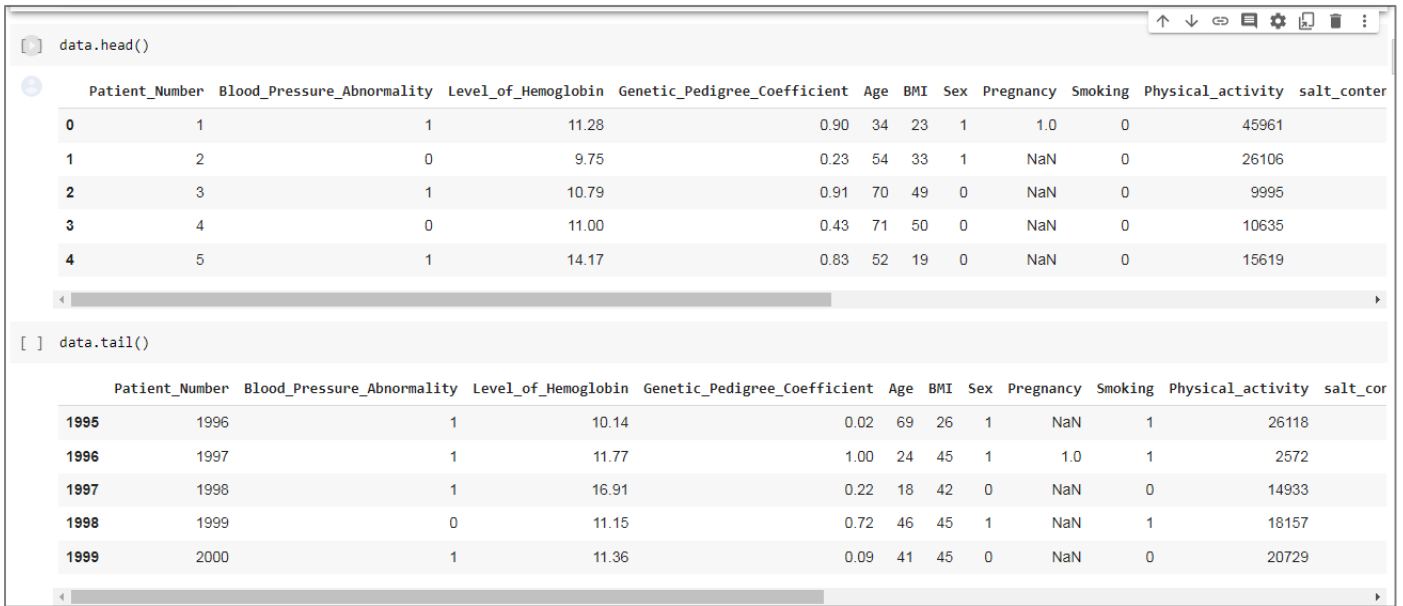
The Python Pillow library is a fork of an older library called PIL. PIL stands for Python Imaging Library, and it's the original library that enabled Python to deal with images.

Methodology



Dataset

In this given dataset their column name and their information:



The screenshot shows a Jupyter Notebook interface with two code cells. The first cell contains `data.head()` and displays the first five rows of a dataset. The second cell contains `data.tail()` and displays the last five rows of the dataset. The dataset has 15 columns: Patient_Number, Blood_Pressure_Abnormality, Level_of_Hemoglobin, Genetic_Pedigree_Coefficient, Age, BMI, Sex, Pregnancy, Smoking, Physical_activity, and salt_content_in_the_diet.

	Patient_Number	Blood_Pressure_Abnormality	Level_of_Hemoglobin	Genetic_Pedigree_Coefficient	Age	BMI	Sex	Pregnancy	Smoking	Physical_activity	salt_content_in_the_diet
0	1	1	11.28	0.90	34	23	1	1.0	0	45961	
1	2	0	9.75	0.23	54	33	1	NaN	0	26106	
2	3	1	10.79	0.91	70	49	0	NaN	0	9995	
3	4	0	11.00	0.43	71	50	0	NaN	0	10635	
4	5	1	14.17	0.83	52	19	0	NaN	0	15619	

	Patient_Number	Blood_Pressure_Abnormality	Level_of_Hemoglobin	Genetic_Pedigree_Coefficient	Age	BMI	Sex	Pregnancy	Smoking	Physical_activity	salt_content_in_the_diet
1995	1996	1	10.14	0.02	69	26	1	NaN	1	26118	
1996	1997	1	11.77	1.00	24	45	1	1.0	1	2572	
1997	1998	1	16.91	0.22	18	42	0	NaN	0	14933	
1998	1999	0	11.15	0.72	46	45	1	NaN	1	18157	
1999	2000	1	11.36	0.09	41	45	0	NaN	0	20729	

Figure 10

In this data set we have to check if there is null value or outlier are in the set of the data. If there is an outlier or null value exist than remove that value or replace with mean, median or mod

```
[ ] data.shape

(2000, 15)

[ ] data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 15 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                ---
0   Patient_Number                        2000 non-null   int64
1   Blood_Pressure_Abnormality            2000 non-null   int64
2   Level_of_Hemoglobin                  2000 non-null   float64
3   Genetic_Pedigree_Coefficient          1908 non-null   float64
4   Age                                   2000 non-null   int64
5   BMI                                   2000 non-null   int64
6   Sex                                   2000 non-null   int64
7   Pregnancy                             442 non-null    float64
8   Smoking                               2000 non-null   int64
9   Physical_activity                     2000 non-null   int64
10  salt_content_in_the_diet               2000 non-null   int64
11  alcohol_consumption_per_day           1758 non-null   float64
12  Level_of_Stress                       2000 non-null   int64
13  Chronic_kidney_disease                2000 non-null   int64
14  Adrenal_and_thyroid_disorders         2000 non-null   int64
dtypes: float64(4), int64(11)
memory usage: 234.5 KB
```

Figure 11

Implementation detail

To fit data in machine learning model we have to split data in two part one is training and second one is the test. After that we can apply various type of machine learning model on this split data.

Step for fitting the model

- a. Split the data in train and test.
- b. After that define x value and y value on the data.
- c. Select a regression model and their parameter that you want to apply on the data.
- d. After that fit training data of x and y in the model.
- e. For predict the value of the given data and put that x part test data in the predict function if the selected model.

We create a code of the data implement from the fryers API through the function of the data and give set criteria to full file the data and fetch the live data

```
root=Tk()
root.geometry("1000x1000")
root.resizable(0,0)
root.configure(bg="lavender")
root.title("Nutrients by Megha Lalwani")
# con=mysql.connector.connect(host='localhost',
#                               user='root',
#                               password='siddharth',
#                               database='nutrients')
# cur=con.cursor()

conn = sqlite3.connect("health_spy.db")
cur = conn.cursor()

def b():
    f=Frame(bg="lightblue")
    f.place(x=0,y=0,width=1000,height=1000)
    vb=Label(f,text="SIGN UP",bg="black",fg="red",font=("",40))
    vb.place(x=0,y=0,height=80,width=1000)
    un1=Label(f,text="Enter Name:",font=("arial",15),bg="lightblue")
    un1.place(x=0,y=100)
    e1=Entry(f,font=("arial",15))
    e1.place(x=150,y=100)

    un2=Label(f,text="Enter Email i'd:",font=("arial",15),bg="lightblue")
    un2.place(x=0,y=150)
    e2=Entry(f,font=("arial",15))
    e2.place(x=150,y=150)
```

Figure 12

After we enter the home page of the API and enter the id and Password of the user and enter the page that generates the Access token for the user.

Nutrients

SIGN UP

Enter Name:

Enter Email i'd:

Enter Age:

Enter Gender:

Enter Phone No.

Enter Password:

sign up

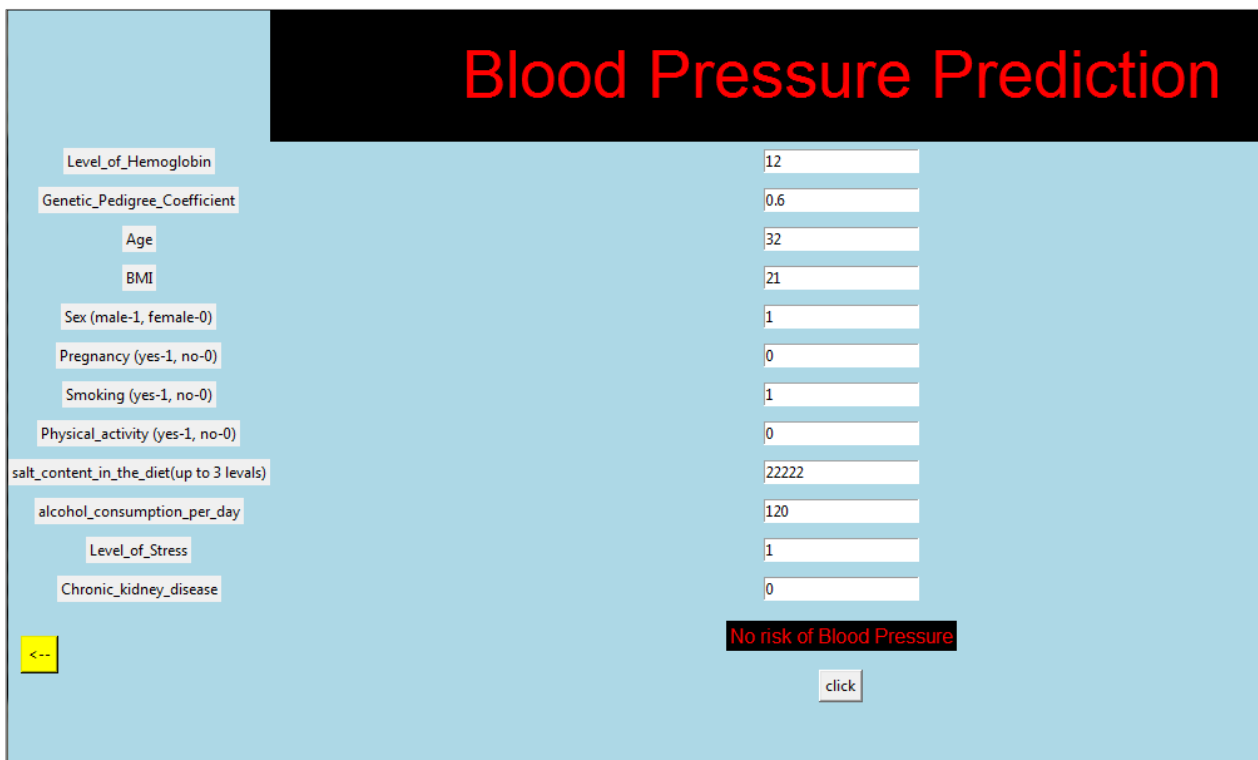
Figure 13



Figure 14

Result

Front page of blood pressure prediction



The form is titled "Blood Pressure Prediction" in red text on a black background. It contains a list of input fields on the left and their corresponding values on the right. The inputs are: Level_of_Hemoglobin (12), Genetic_Pedigree_Coefficient (0.6), Age (32), BMI (21), Sex (male-1, female-0) (1), Pregnancy (yes-1, no-0) (0), Smoking (yes-1, no-0) (1), Physical_activity (yes-1, no-0) (0), salt_content_in_the_diet(up to 3 levals) (22222), alcohol_consumption_per_day (120), Level_of_Stress (1), and Chronic_kidney_disease (0). A yellow button with "<--" is at the bottom left. A red box with "No risk of Blood Pressure" is at the bottom right, with a "click" button below it.

Input Field	Value
Level_of_Hemoglobin	12
Genetic_Pedigree_Coefficient	0.6
Age	32
BMI	21
Sex (male-1, female-0)	1
Pregnancy (yes-1, no-0)	0
Smoking (yes-1, no-0)	1
Physical_activity (yes-1, no-0)	0
salt_content_in_the_diet(up to 3 levals)	22222
alcohol_consumption_per_day	120
Level_of_Stress	1
Chronic_kidney_disease	0

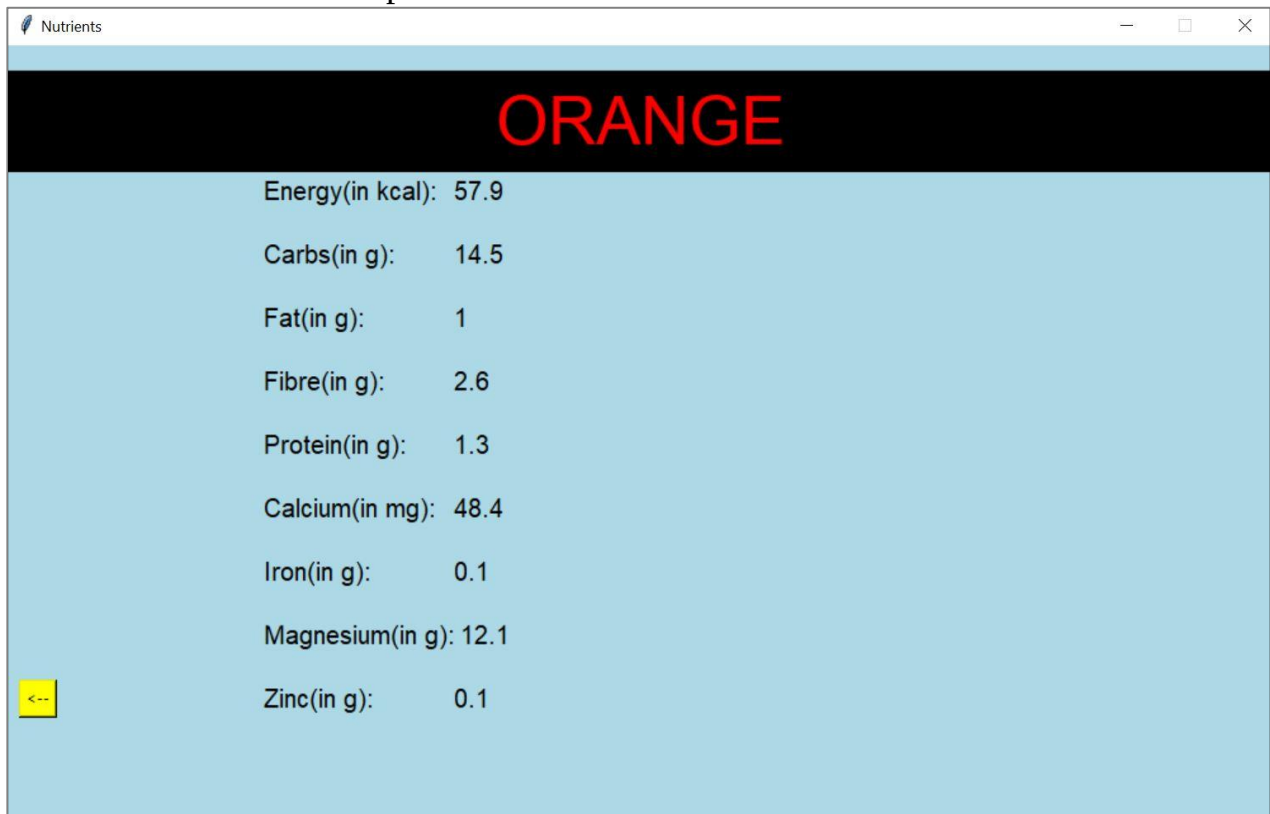
<--

No risk of Blood Pressure

click

Figure 15

Other Nutrients and health tips



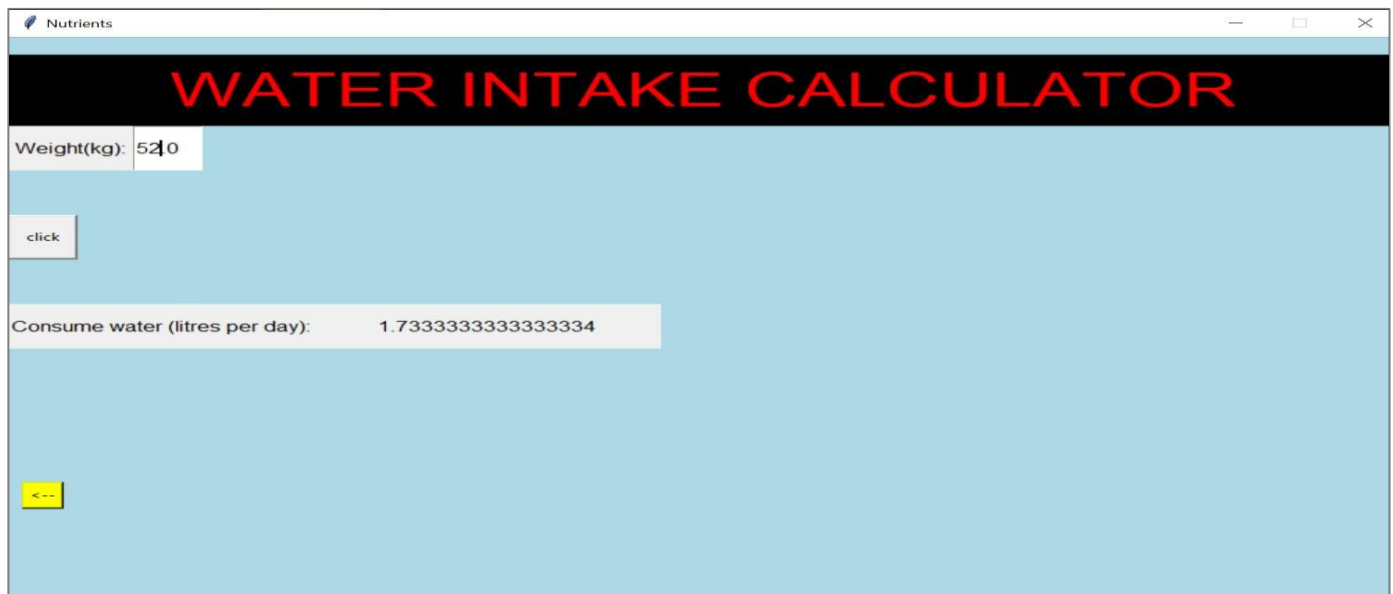
The window is titled "Nutrients" and has a black header with "ORANGE" in red text. It displays a list of nutrients and their values: Energy(in kcal): 57.9, Carbs(in g): 14.5, Fat(in g): 1, Fibre(in g): 2.6, Protein(in g): 1.3, Calcium(in mg): 48.4, Iron(in g): 0.1, Magnesium(in g): 12.1, and Zinc(in g): 0.1. A yellow button with "<--" is at the bottom left.

Nutrient	Value
Energy(in kcal)	57.9
Carbs(in g)	14.5
Fat(in g)	1
Fibre(in g)	2.6
Protein(in g)	1.3
Calcium(in mg)	48.4
Iron(in g)	0.1
Magnesium(in g)	12.1
Zinc(in g)	0.1

<--

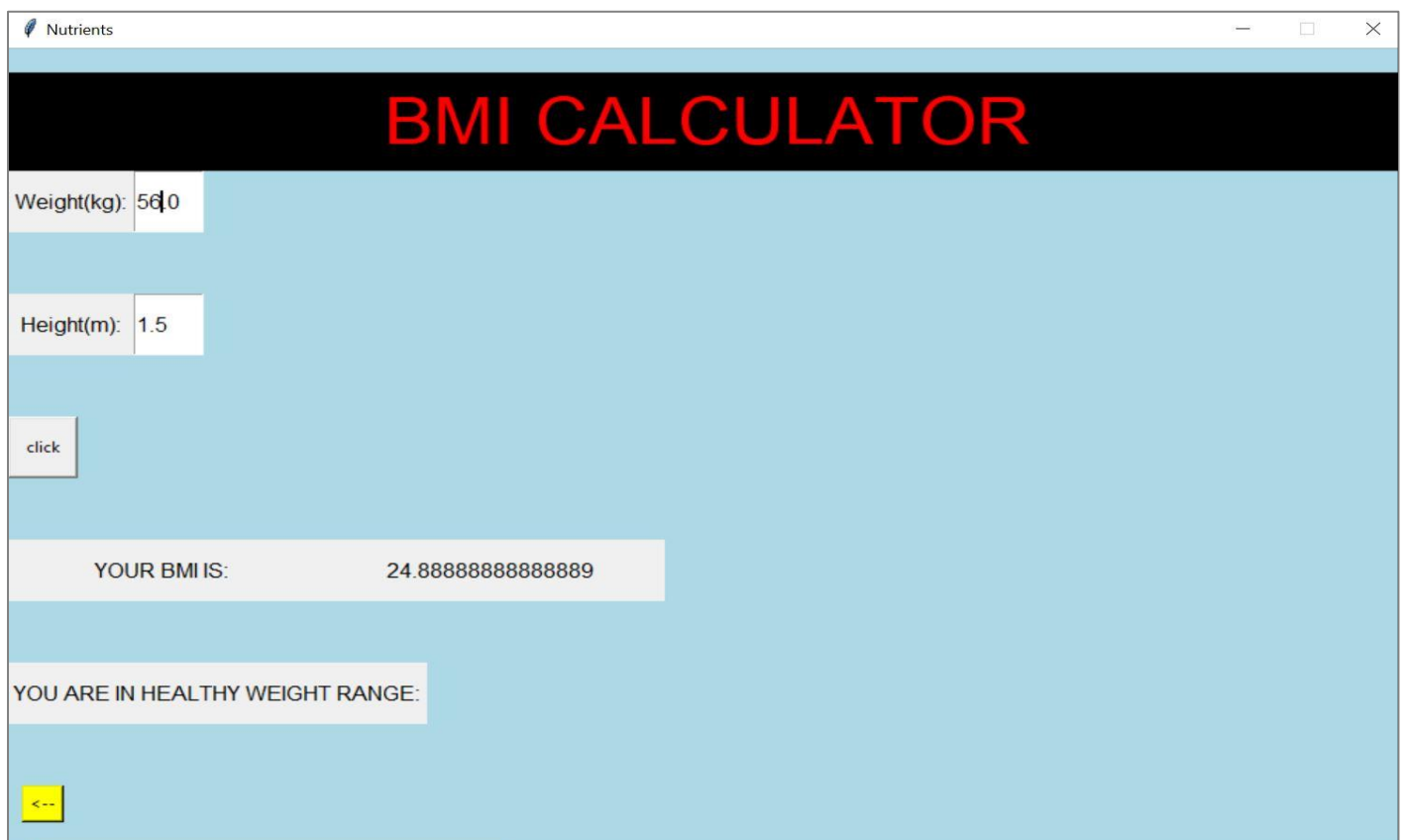
Figure 16

BMI and Water intake calculate



The screenshot shows a web application window titled "Nutrients". The main heading is "WATER INTAKE CALCULATOR" in large red letters. Below the heading, there is a text input field for "Weight(kg):" with the value "52.0". A "click" button is positioned below the weight input. Further down, a text input field for "Consume water (litres per day):" displays the value "1.7333333333333334". At the bottom left, there is a yellow button with the text "<--".

Figure 17



The screenshot shows a web application window titled "Nutrients". The main heading is "BMI CALCULATOR" in large red letters. Below the heading, there are two text input fields: "Weight(kg):" with the value "52.0" and "Height(m):" with the value "1.5". A "click" button is positioned below the height input. Further down, a text input field displays "YOUR BMI IS: 24.88888888888889". Below that, another text input field displays "YOU ARE IN HEALTHY WEIGHT RANGE:". At the bottom left, there is a yellow button with the text "<--".

Figure 18

References

- [The genetics of blood pressure and hypertension: the role of rare variation - PubMed \(nih.gov\)](#)
- [What Is Normal Blood Pressure Range by Age? Adults, Children & Charts \(emedicinehealth.com\)](#)
- [Low Blood Pressure During Pregnancy: What You Should Know \(healthline.com\)](#)
- [What is salt and how does it affect our blood pressure? | Eufic](#)
- [Blood Pressure After Exercising: What's Normal, Seeking Help, and Safety \(healthline.com\)](#)
- [Stress and high blood pressure: What's the connection? - Mayo Clinic](#)
- <https://www.tutorialspoint.com/index.htm>